Su Jiaji

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Education

Jan 2024 | **Doctor of Philosophy in Statistics**
Department of Statistics & Data Science
National University of Singapore, Singapore
Thesis Advisor: Dr. YAO Zhigang

Jun 2018 | **Bachelor of Natural Science in Statistics**
School of Mathematical Science
Zhejiang University, Hangzhou, China

Professional Experience

Mar 2026 – Current | **Hongshen Young Faculty**
College of Mathematics and Statistics
Chongqing University, Chongqing, China

Oct 2023 – Feb 2026 | **Research Fellow** | Advisor: Dr. YAO Zhigang
Department of Statistics & Data Science
National University of Singapore, Singapore

Aug 2022 – Oct 2023 | **Student Research Assistant** | Advisor: Dr. YAO Zhigang
Department of Statistics & Data Science
National University of Singapore, Singapore

Selected Ongoing Projects

Since Jun 2025 | **Manifold Fitting for Biological Sequence Embeddings**
- Extends our manifold fitting framework to high-dimensional embeddings produced by Transformer-based DNA/RNA language models.
- Identifies low-dimensional structures that coherently organise sequences sharing biological functions, providing geometric insight into the "sequence language space."
- Incorporates the fitted manifolds into classification, clustering, and other downstream tasks, enhancing accuracy and interpretability over raw embeddings.

Since Jan 2024 | **Low-dimension Structure Analysis Based on UK Biobank**
- Applies manifold-fitting methods to large-scale biomedical data, with a focus on the UK Biobank.
- Aims to uncover low-dimensional structures within high-dimensional population health data.
- Leverages these structures to improve population stratification and refine medical risk assessments.

Preprints & Publications

*Equal Contribution; #Correspondence

Accepted:

*Li, B., *Su, J., *Lin, R., #Yau, S.-T., & #Yao, Z. (2025). Manifold fitting reveals metabolomic heterogeneity and disease associations in UK biobank populations. *Proceedings of the National Academy of Sciences*, *122*(22), e2500001122.

#*Yao, Z., *Su, J., & #*Yau, S.-T. (2024). Manifold fitting with cycleGAN. *Proceedings of the National Academy of Sciences*, *121*(5), e2311436121.

Su, J., #Yao, Z., Li, C., & Zhang, Y. (2023). A statistical approach to estimating adsorption-isotherm parameters in gradient-elution preparative liquid chromatography. *The Annals of Applied Statistics*, *17*(4), 3476–3499.

PREPRINTS:

Su, J. & #Yao, Z. (2025). Principal decomposition with nested submanifolds. *arXiv preprint arXiv:2502.10010*. (*Biometrika RR*).

#Yao, Z., Su, J., Li, B., & Yau, S.-T. (2023). Manifold fitting. *arXiv preprint arXiv:2304.07680*. (*JMLR under review*).

TALKS & PRESENTATIONS

MAY 2025 Beijing	YMSC STATISTICAL SEMINAR *Yau Mathematical Sciences Center, Tsinghua University*
JAN 2025 Shanghai & Sanya	THE SECOND SYMPOSIUM OF GEOMETRY AND STATISTICS IN CHINA *Shanghai Institute for Mathematics and Interdisciplinary Sciences* & *The Tsinghua Sanya International Mathematics Forum*
OCT 2024 Singapore	INTERACTIONS OF STATISTICS AND GEOMETRY (ISAG) II *Institute for Mathematical Sciences, National University of Singapore*
JUN 2024 Shanghai	SUMMER SEMINAR SERIES: STATISTICS + GEOMETRY + X *Shanghai Institute for Mathematics and Interdisciplinary Sciences*
JUL 2023 Beijing	TALK: 'ESTIMATING ADSORPTION-ISOTHERM PARAMETERS' *School of Mathematics and Statistics, Beijing Institute of Technology*

VISITING

JUN 2024 – JUL 2024	VISITING RESEARCHER Shanghai Institute for Mathematics and Interdisciplinary Sciences, China
DEC 2024 – JAN 2025	VISITING RESEARCHER Shanghai Institute for Mathematics and Interdisciplinary Sciences, China

TEACHING

Teaching assistant at National University of Singapore:

Semester	Courses	
2021–22 Sem 2	ST2334	Probability and Statistics
	DSA1101	Introduction to Data Science
	ST2137	Statistical Computing and Programming
2021–22 Sem 1	ST5213	Advanced Categorical Data Analysis
	DSA1101	Introduction to Data Science
2020–21 Sem 2	ST2334	Probability and Statistics
	ST3232	Design and Analysis of Experiments
2020–21 Sem 1	ST2334	Probability and Statistics
	ST5225	Statistical Analysis of Networks
2019–20 Sem 2	ST2131	Probability
	ST3239	Survey Methodology
2019–20 Sem 1	ST4231	Computer Intensive Statistical Methods
2018–19 Sem 2	ST2131	Probability

Last updated: March 24, 2026